Splitting Idempotents via Interpolation in Multiplicative Linear Logic

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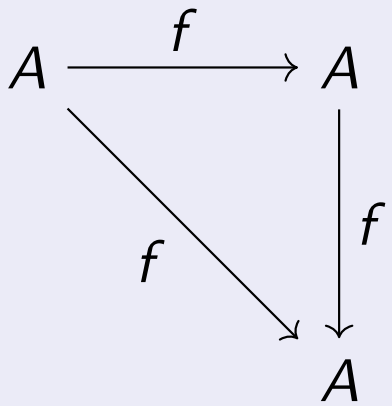
Nancy, 21 October 2025





(Split) Idempotents

Definition: Idempotent

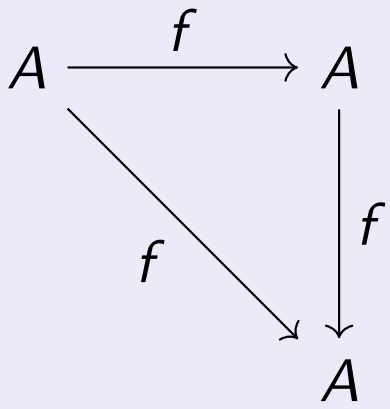


Why studying idempotents?

Gives a better understanding
(*e.g.* in group theory, λ-calculus)

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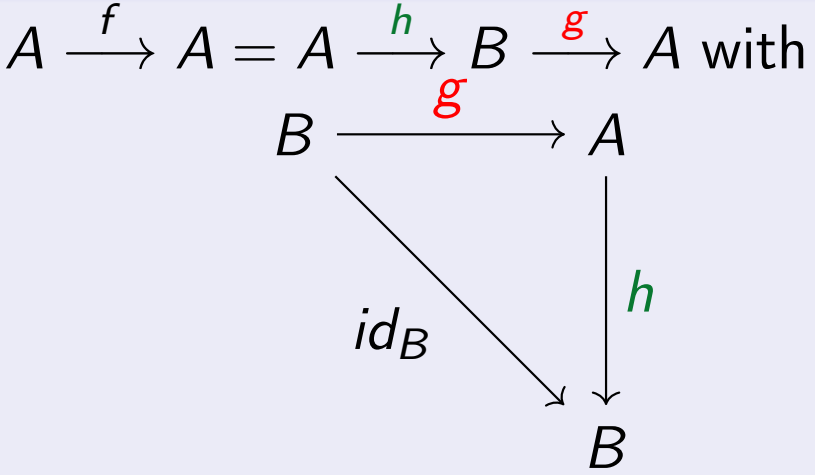
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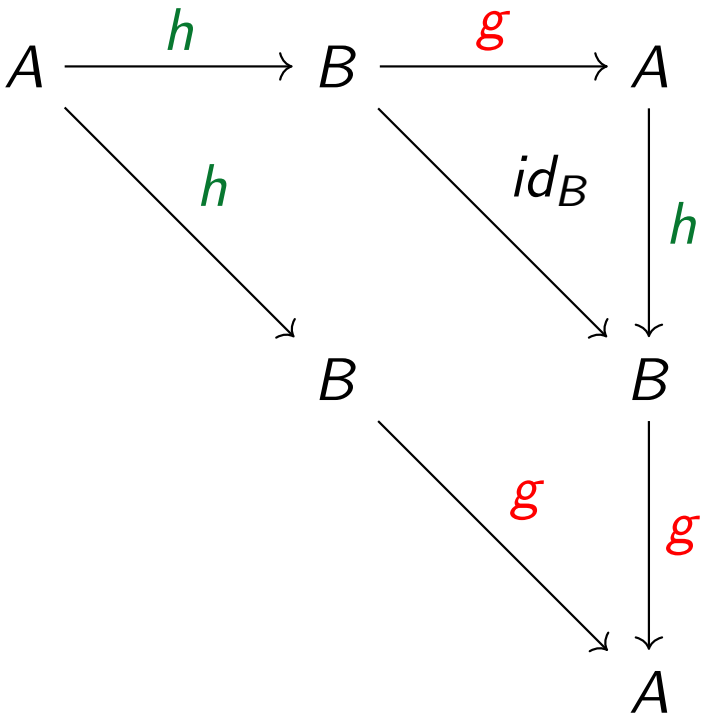


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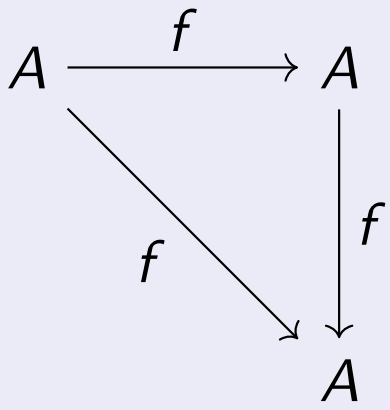
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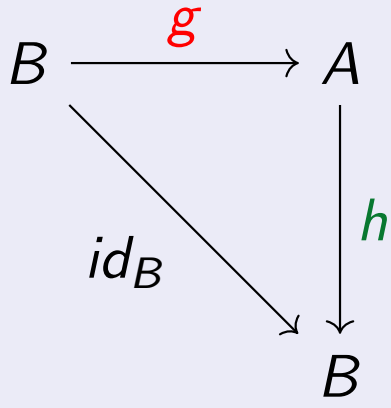
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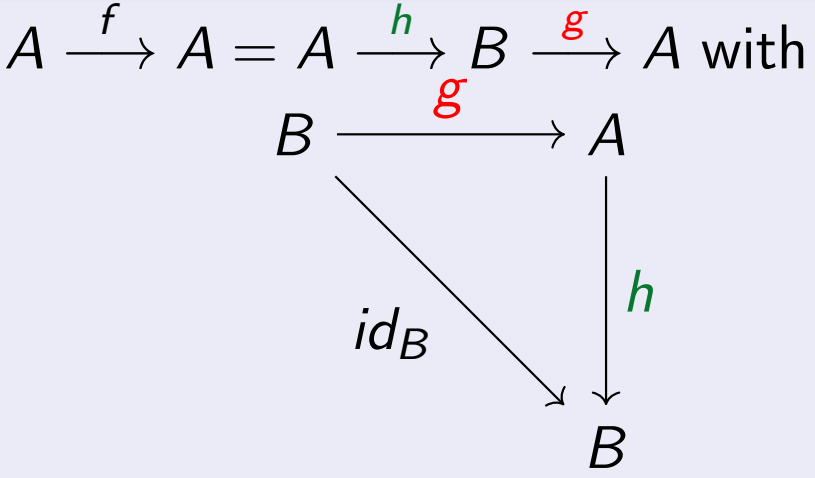
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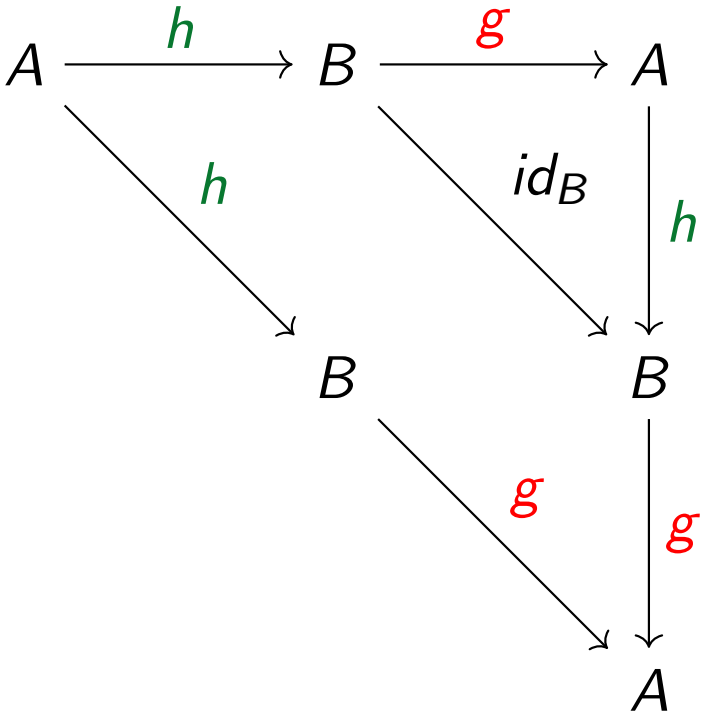
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Definition: Section-Retraction



Definition: Split Idempotent





Idempotent Completeness

Are all idempotents split?

If yes, the category is **idempotent complete**

(*a.k.a.* Karoubi complete, Cauchy complete)

Around idempotent completeness

- Nice property, *e.g.* every absolute (co)limit comes from split idempotents **[Borceux and Dejean 1986]**
- idempotents $\approx$ section-retractions
- Some completion procedures to turn a category idempotent complete

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Set Theory ✓ **[folklore]**

$$\text{idempotent } A \xrightarrow{f} A = A \overset{f^{|}}{\twoheadrightarrow} f(A) \overset{\subseteq}{\hookrightarrow} A$$

$$\text{with } f(A) \overset{\subseteq}{\hookrightarrow} A \overset{f^{|}}{\twoheadrightarrow} f(A) : f(a) \mapsto f(f(a)) = f(a)$$

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MLL/free $\star$-autonomous category ✓ **[this talk]**

Proving MLL is Idempotent Complete

Plan

Given a proof τ of $A \vdash A$ that is idempotent:

1. **Find** a formula B and proofs π of $A \vdash B$ and ϕ of $B \vdash A$ such that

$$\tau = \pi \circ \phi$$

2. **Show** that

$$\phi \circ \pi = id_B$$

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We get the difficult 1 by **proof-relevant interpolation [Saurin 2025]**

Theorem: Proof-Relevant Interpolation (reformulated)

Let τ be a proof of $A \vdash A$ in linear logic. There exists a formula B and proofs π of $A \vdash B$ and ϕ of $B \vdash A$ such that $\pi \circ \phi = \tau$.

(+ some properties)

Multiplicative Linear Logic

Formulas

$$A, B \ ::= \ | \ X^- \ | \ X^+ \ | \ A \parr B \ | \ A \otimes B \ | \ \bot \ | \ 1$$

Involutive Negation

$$(X^-)^\bot = X^+ \quad (X^+)^\bot = X^-$$

$$(A \parr B)^\bot = A^\bot \otimes B^\bot \quad (A \otimes B)^\bot = A^\bot \parr B^\bot \quad \bot^\bot = 1 \quad 1^\bot = \bot$$

Rules (one-sided)

$$\frac{}{\vdash X^-, X^+}\ (ax) \qquad \frac{\vdash A^\bot, \Gamma \qquad \vdash A, \Delta}{\vdash \Gamma, \Delta}\ (cut)$$

$$\frac{\vdash A, B, \Gamma}{\vdash A \parr B, \Gamma}\ (\parr) \qquad \frac{\vdash A, \Gamma \qquad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta}\ (\otimes) \qquad \frac{\vdash \Gamma}{\vdash \bot, \Gamma}\ (\bot) \qquad \frac{}{\vdash 1}\ (1)$$

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Axiom-Expansion η **Cut-Elimination** β

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Back to the Plan

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Given a proof τ such that $\dfrac{\overset{\tau}{\vdash A^\perp, A} \quad \overset{\tau}{\vdash A^\perp, A}}{\vdash A^\perp, A}\,(cut) \quad = \quad \tau$

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Given a proof τ such that $\dfrac{\overset{\tau}{\vdash A^\perp, A} \quad \overset{\tau}{\vdash A^\perp, A}}{\vdash A^\perp, A}\,(cut) \quad =_{\beta\eta} \quad \tau$

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Problem

1 is given by interpolation; how to prove 2?

Deciding Equality in MLL

Theorem [Heijltjes and Houston 2016]

MLL proof equivalence is PSPACE-complete.

☹ The general problem is hard ☹ *(the unit $\bot$ moves too freely)*

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Theorem

Two proofs π and ϕ of $\vdash A^\bot, A$ are equivalent if and only if

- *their ax-rules regroup the same atoms; and*
- *their* $\dfrac{\dfrac{}{\vdash 1}\,(1)}{\vdash \bot, 1}\,(\bot)$ *regroup the same* 1 *and* $\bot$.

(Linear complexity)

☺ Our particular case is easy ☺ *(each $\bot$ is stuck below a 1)*

Deciding Equality in MLL

Theorem [Heijltjes and Houston 2016]

MLL proof equivalence is PSPACE-complete.

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Theorem

Two (normal) proofs π and ϕ of $\vdash A^{\bot}, A$ are equivalent if and only if (up to isomorphism)

- *their ax-rules regroup the same (occurrences of) atoms; and*
- *their* $\dfrac{\dfrac{}{\vdash 1}\,(1)}{\vdash \bot, 1}\,(\bot)$ *regroup the same (occurrences of)* 1 *and* $\bot$.

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Fragile

Works in MLL but fails when adding additives, exponentials, or quantifiers

New Objective

Updated Plan

Given a proof τ such that $\dfrac{\overset{\tau}{\vdash A^\perp, A} \quad \overset{\tau}{\vdash A^\perp, A}}{\vdash A^\perp, A}\,(cut) \quad =_{\beta\eta} \quad \tau$

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$$\tau \quad =_{\beta\eta} \quad \frac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,(cut)$$

   with some properties on π and ϕ: each *ax*-rule uses at least one atom from $A^\perp$ or A, and similarly for units (up to changes in the algorithm)

2. Given a *normal form* μ of $\dfrac{\overset{\phi}{\vdash B^\perp, A} \quad \overset{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B}\,(cut)$ **show** it has

   ax-rules between dual atoms of $A^\perp$ and A, and similarly for units
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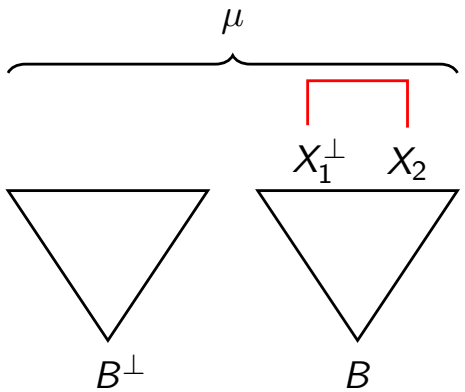
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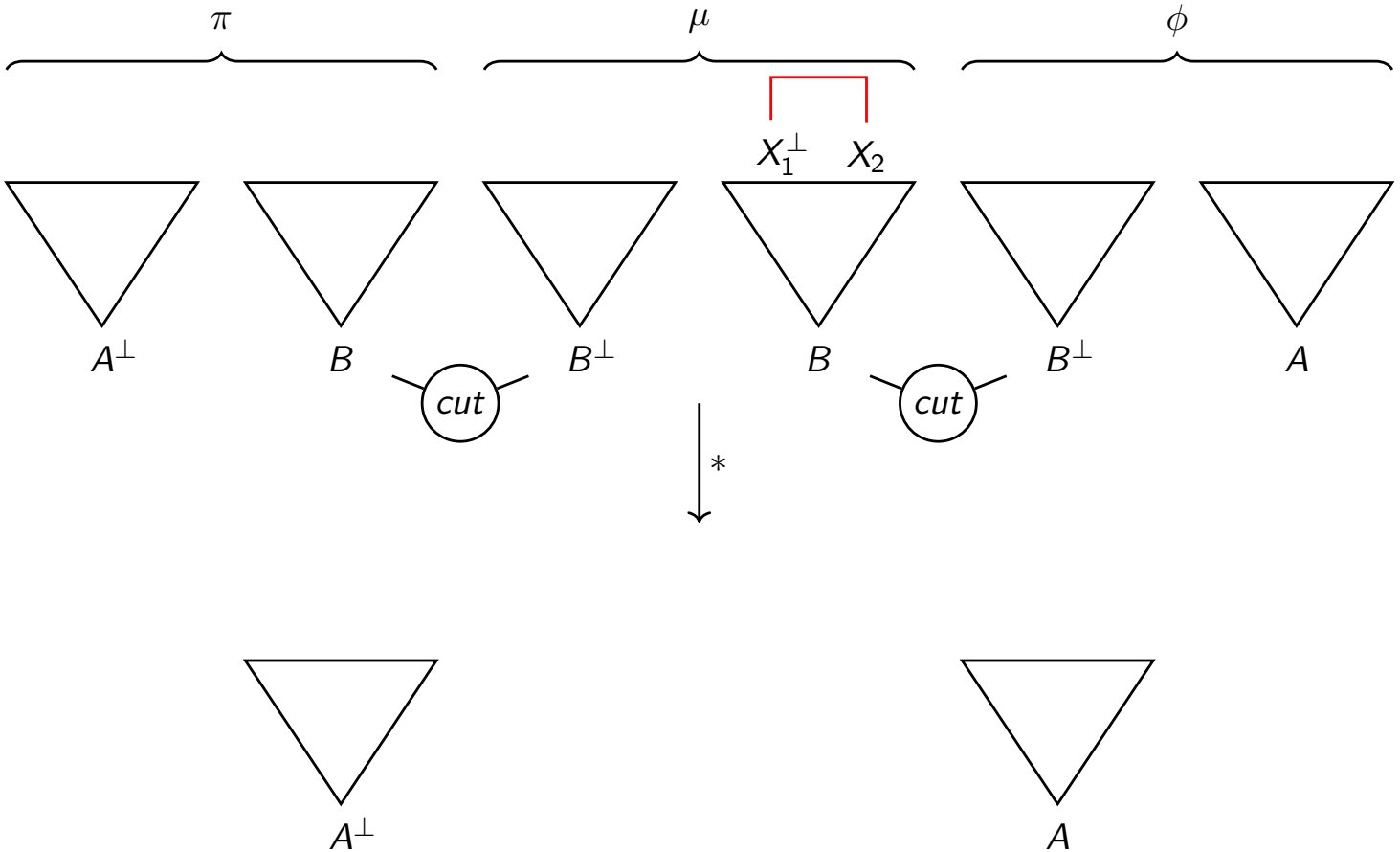
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Atoms and units behave similarly here
$\Longrightarrow$ look only at atoms, graphically: proof-nets / formula-trees!

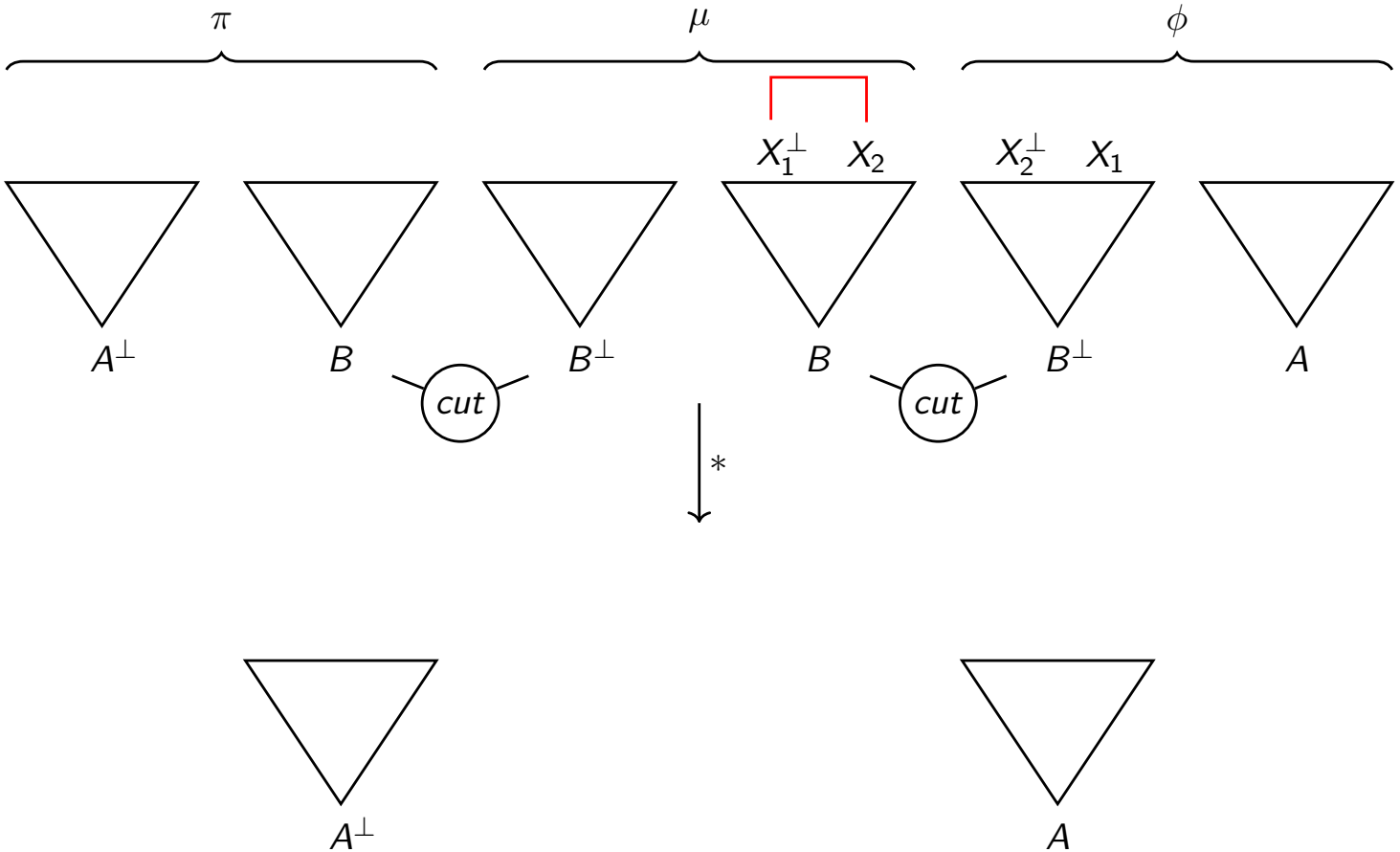
Graphical Axiom-Chasing: No $B - B$ axiom



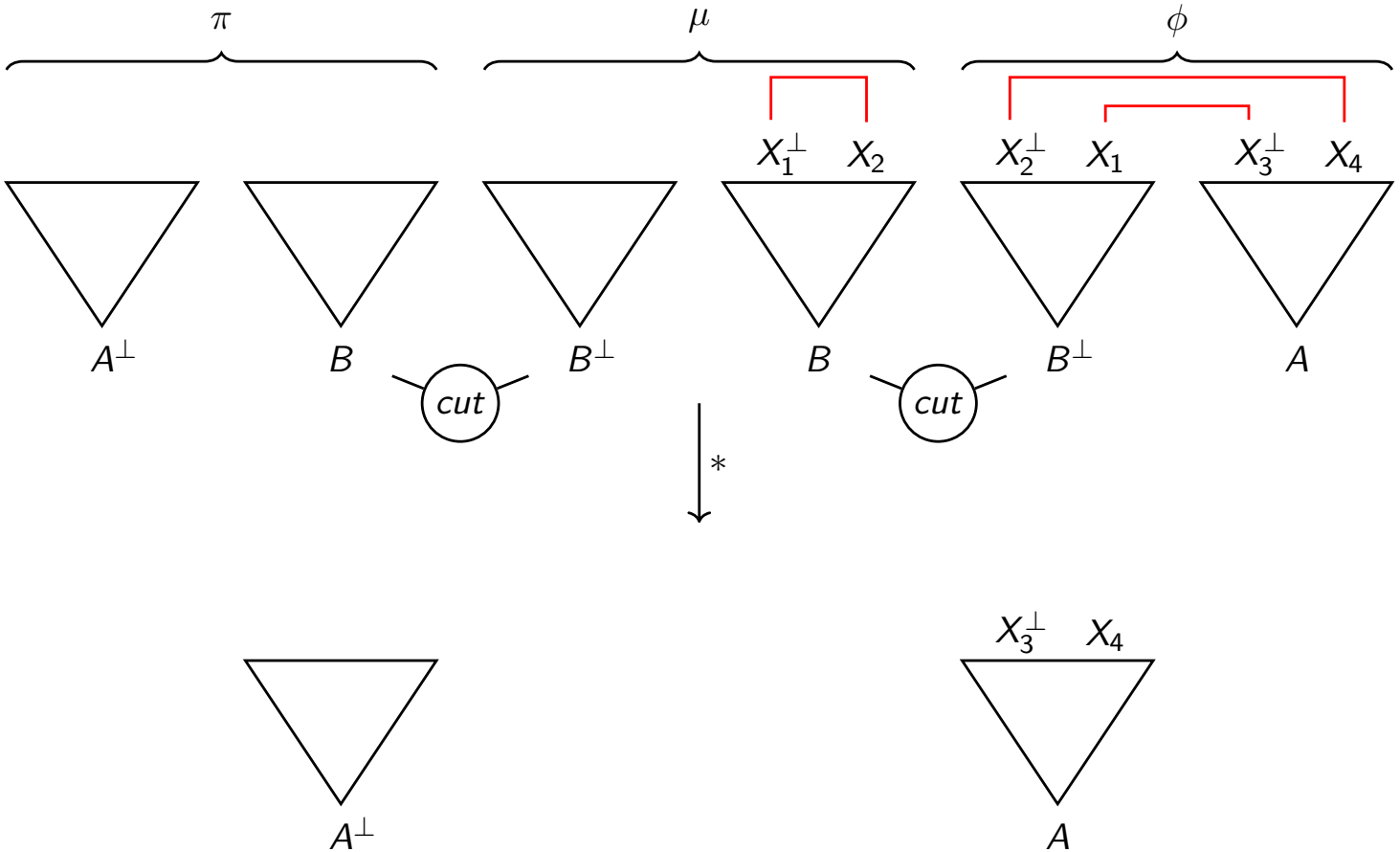
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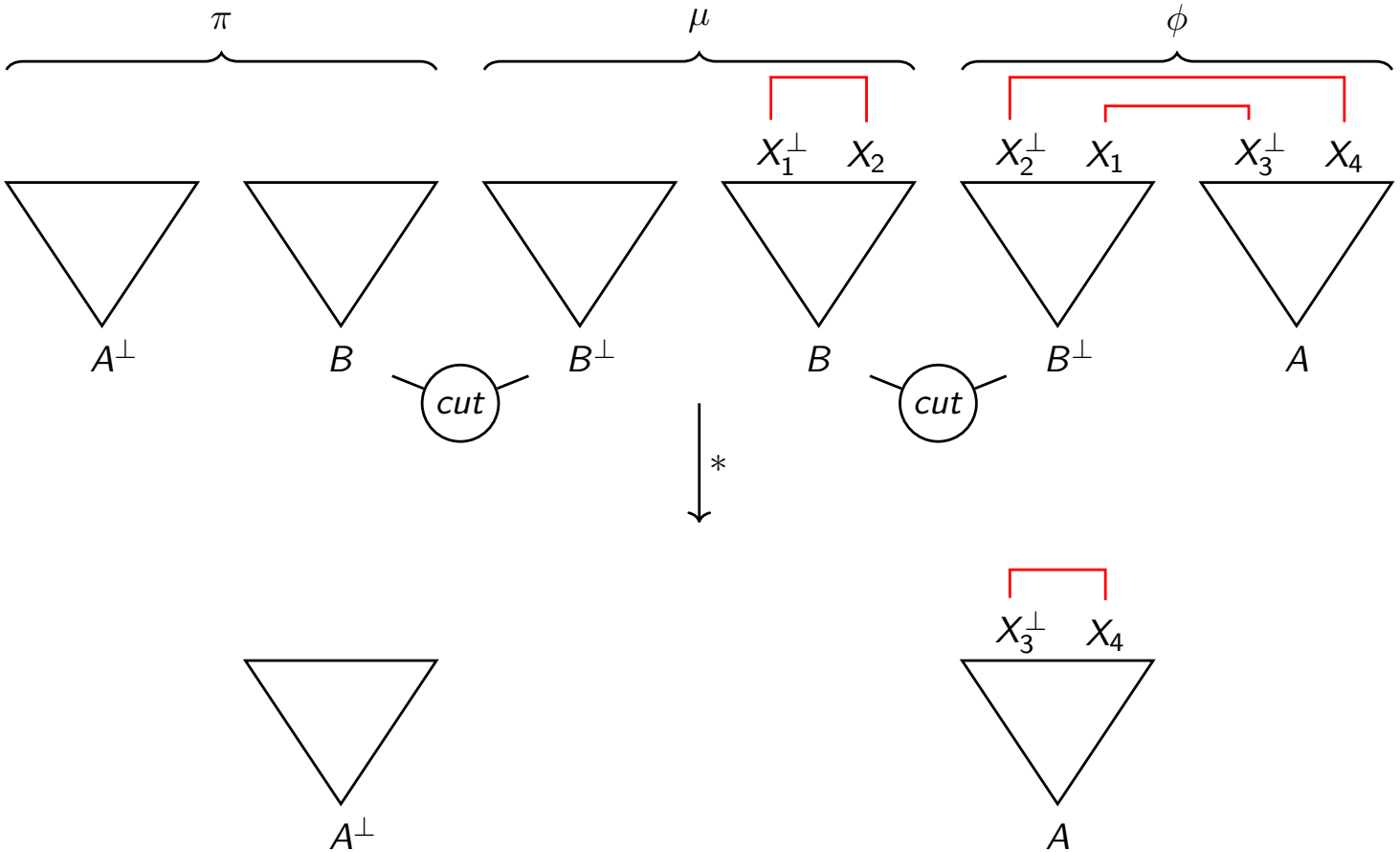
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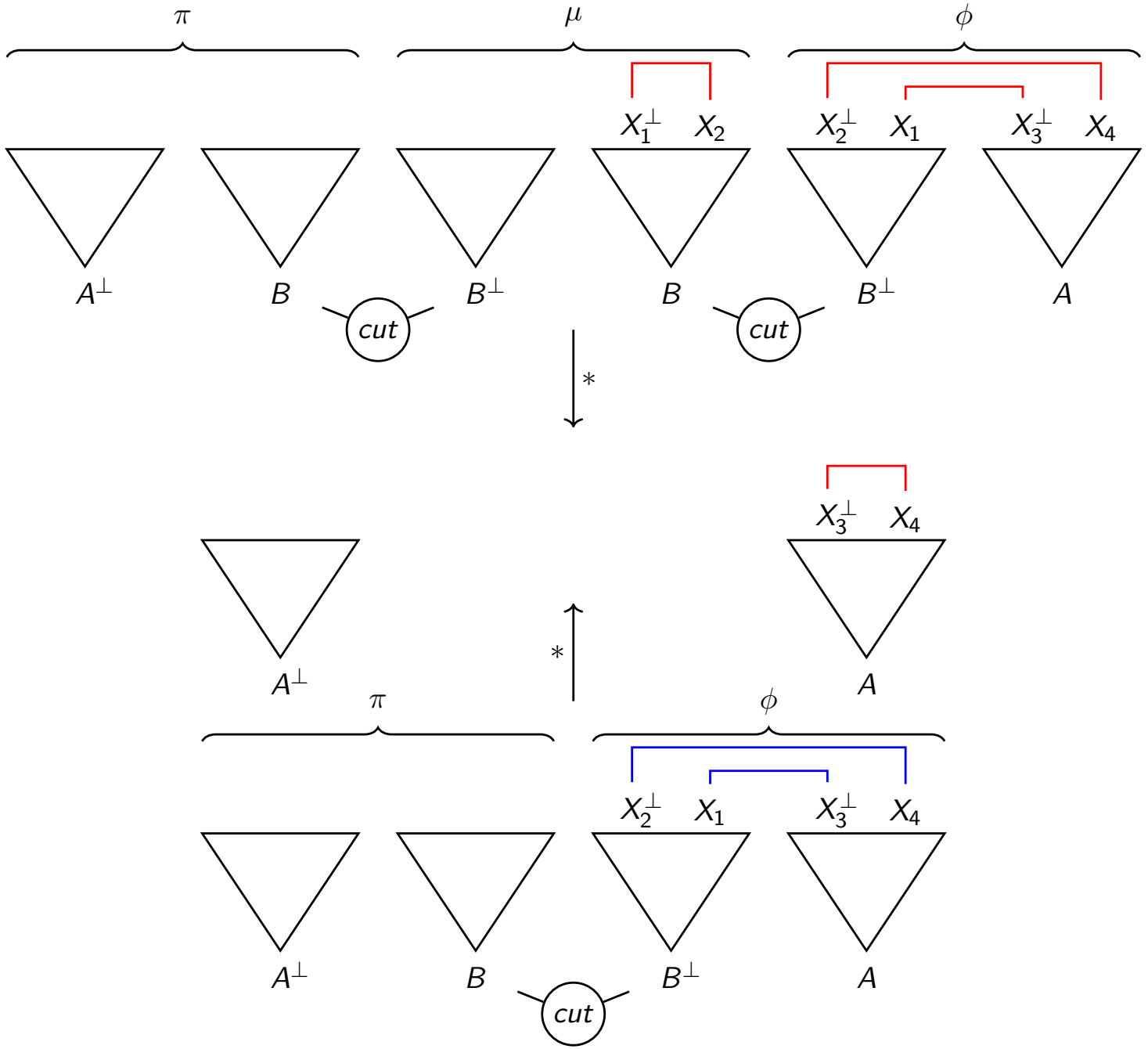
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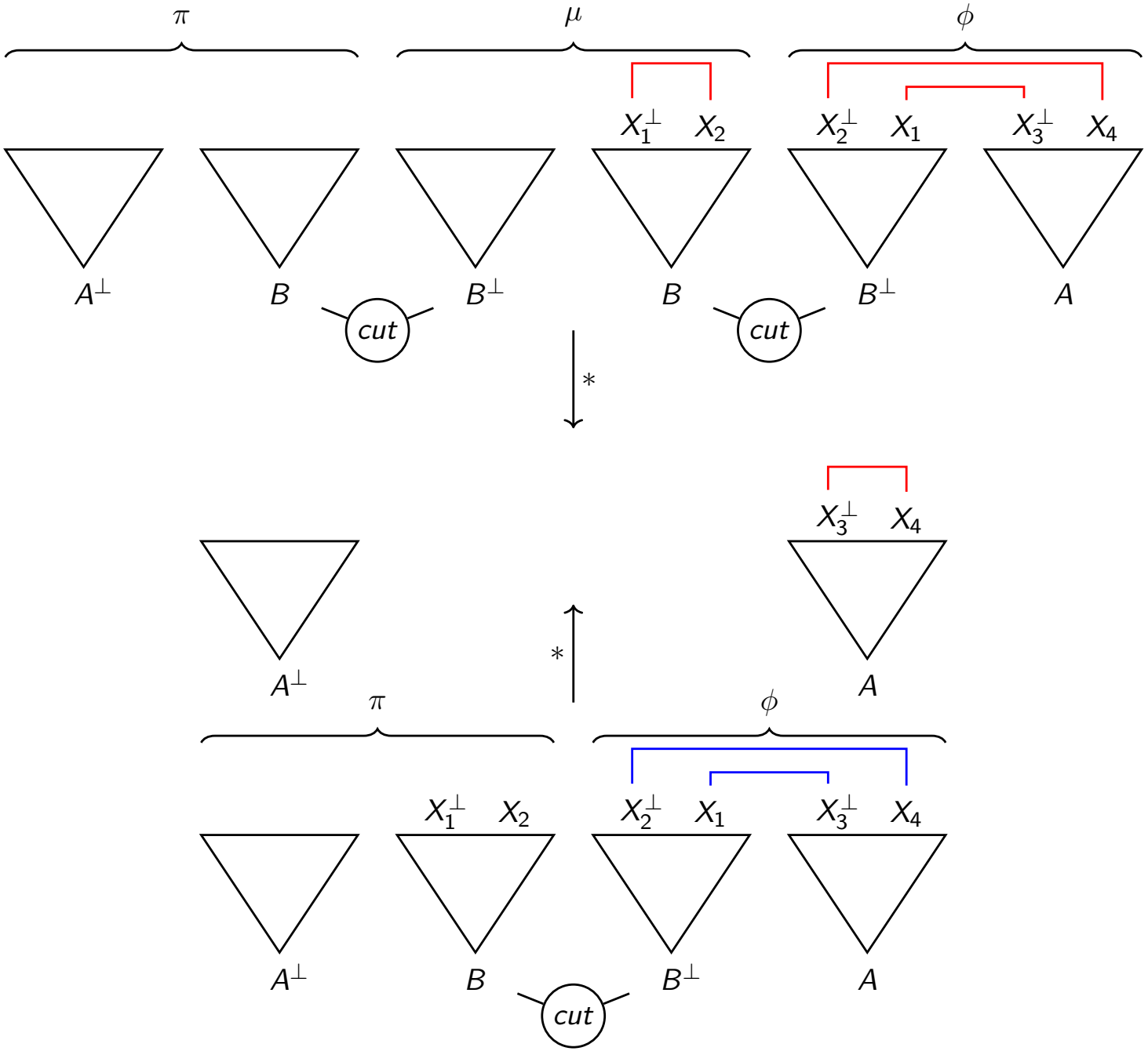
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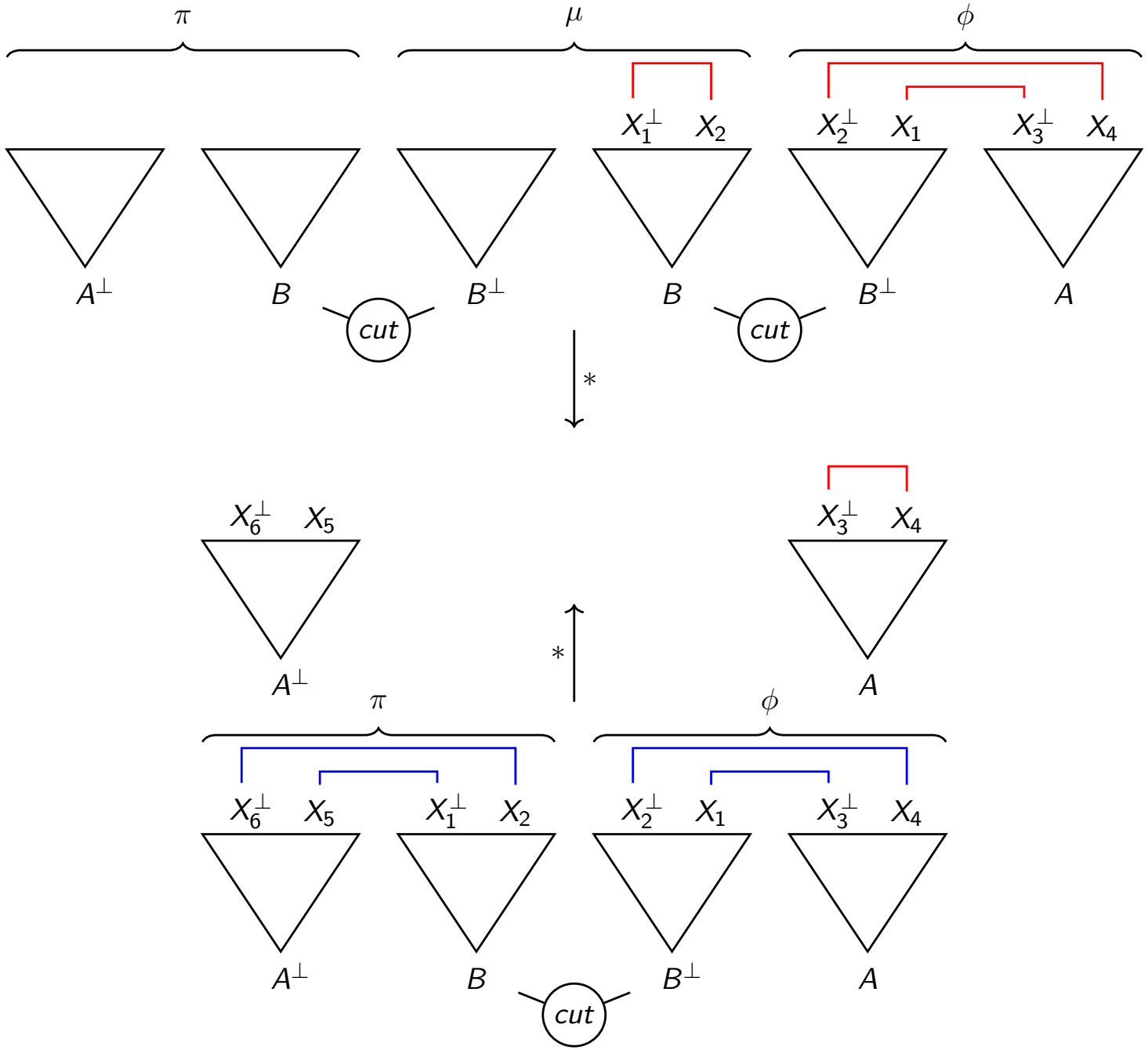
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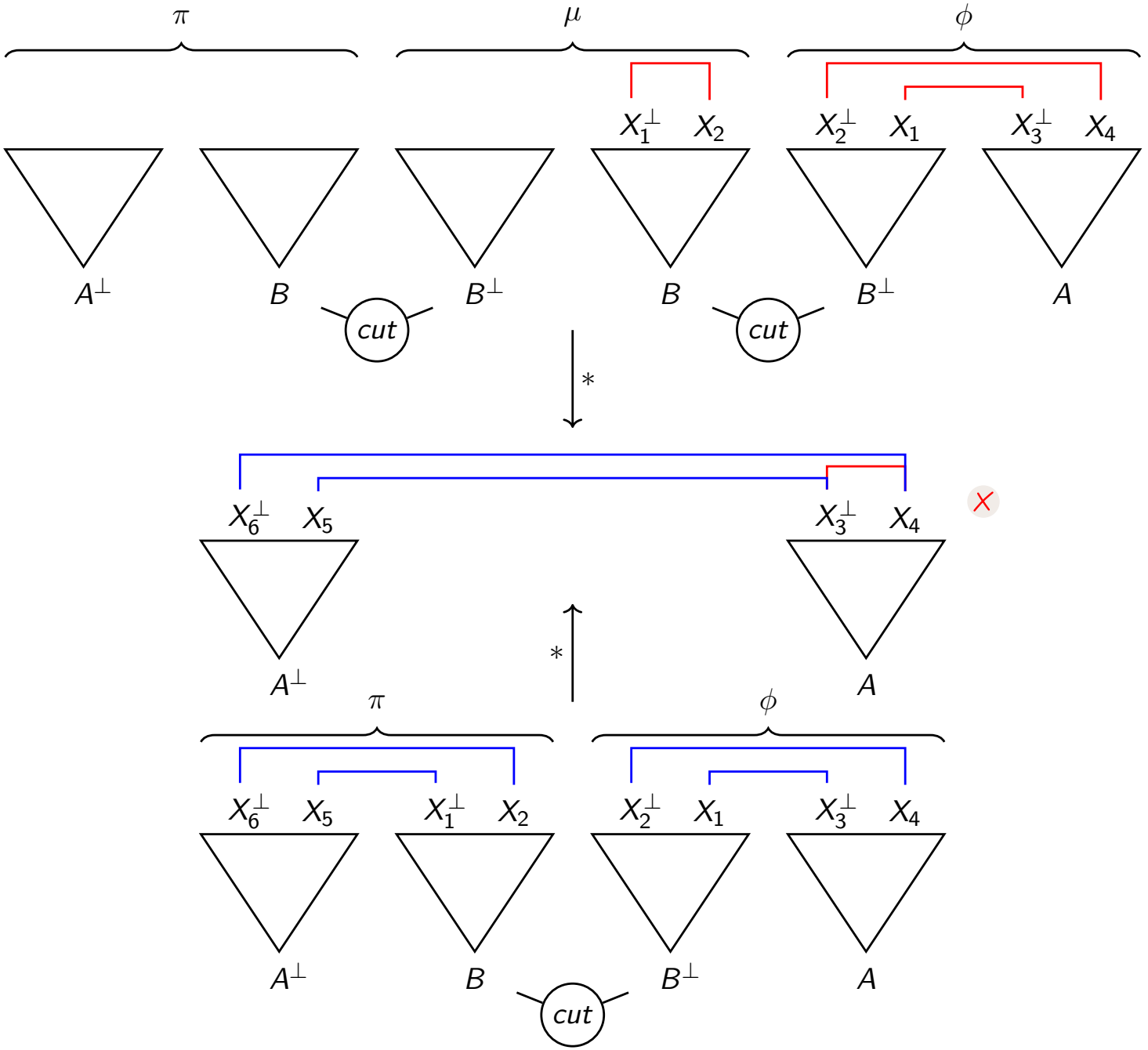
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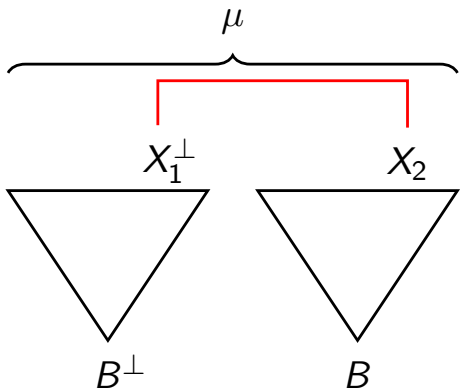
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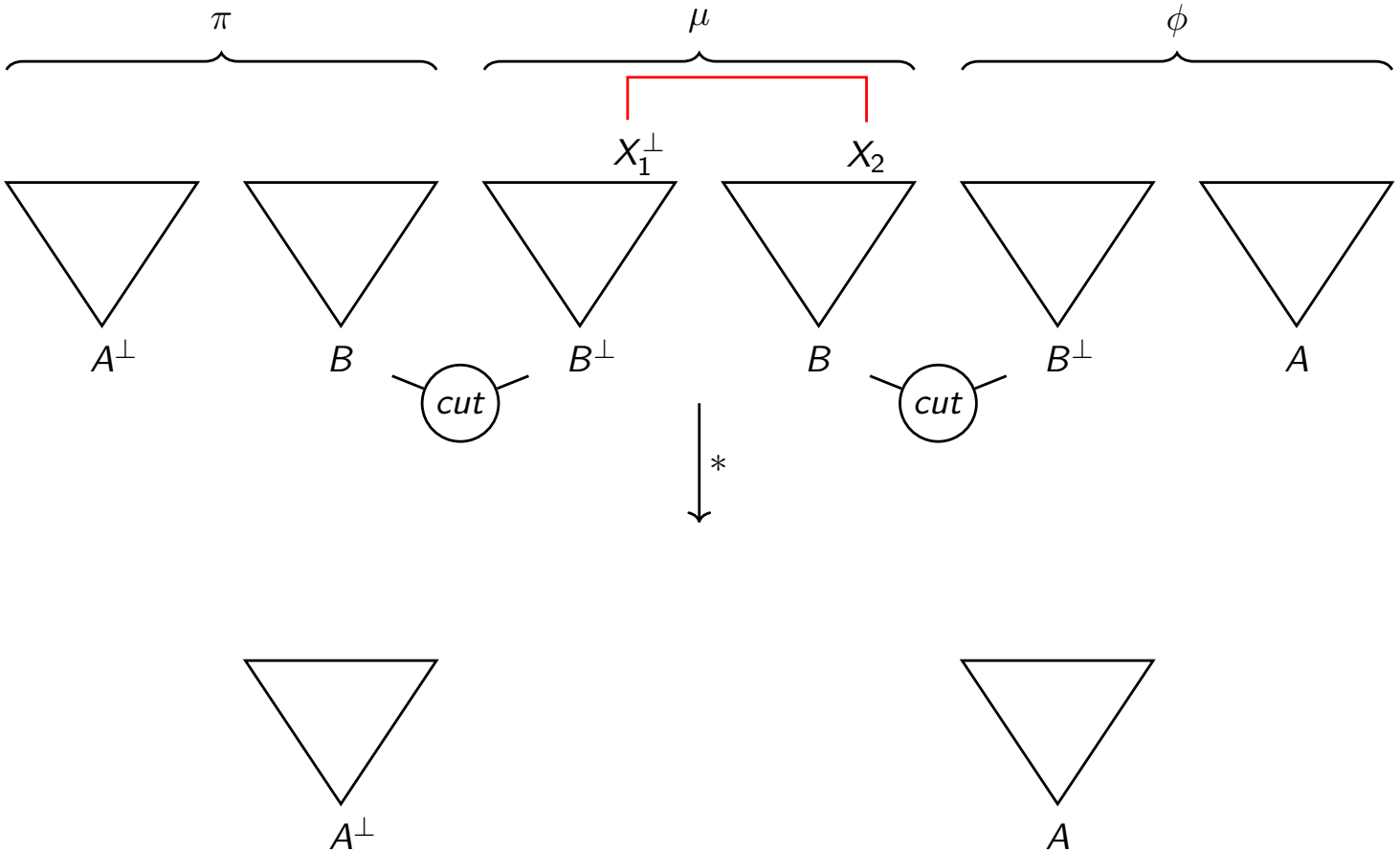
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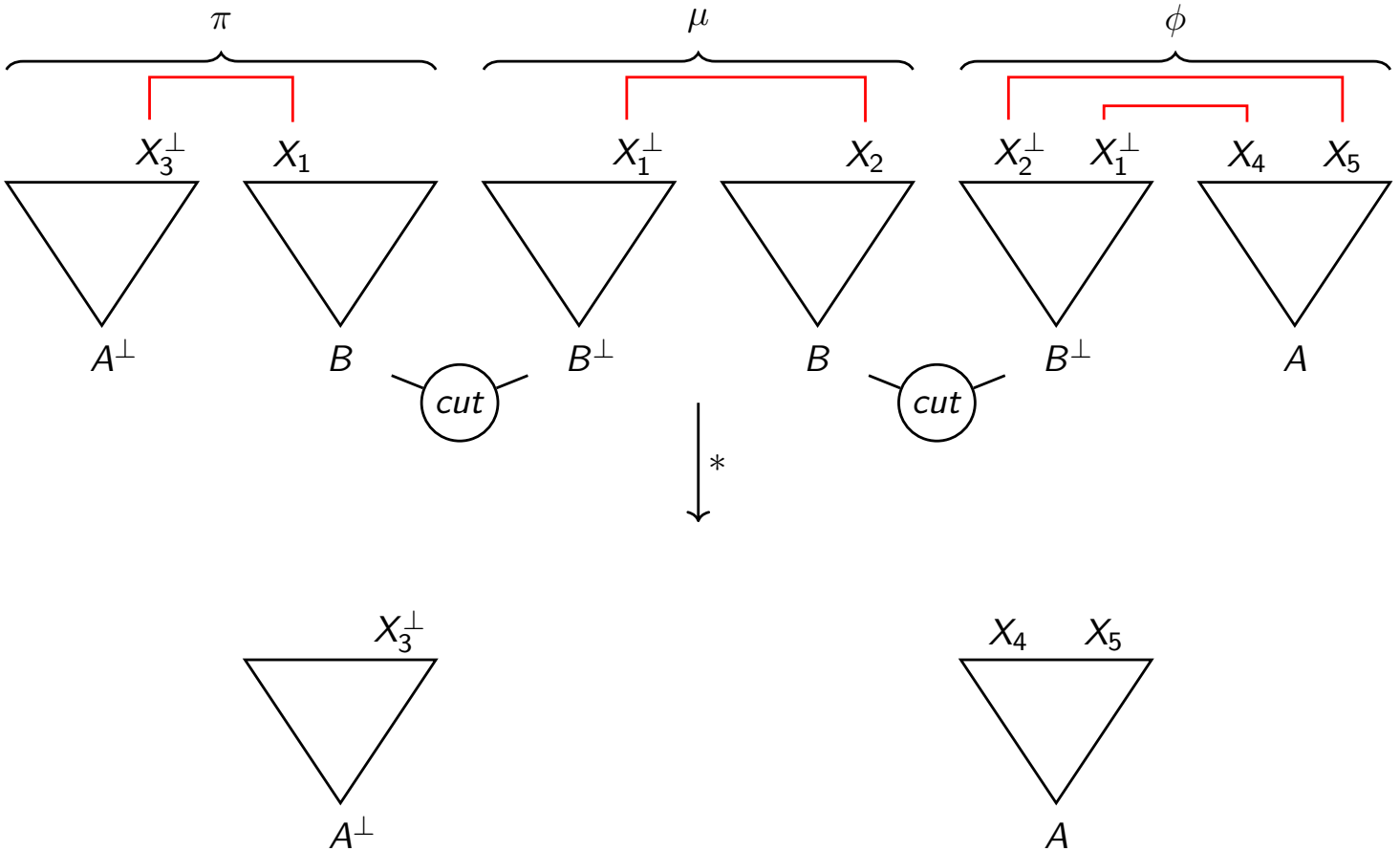
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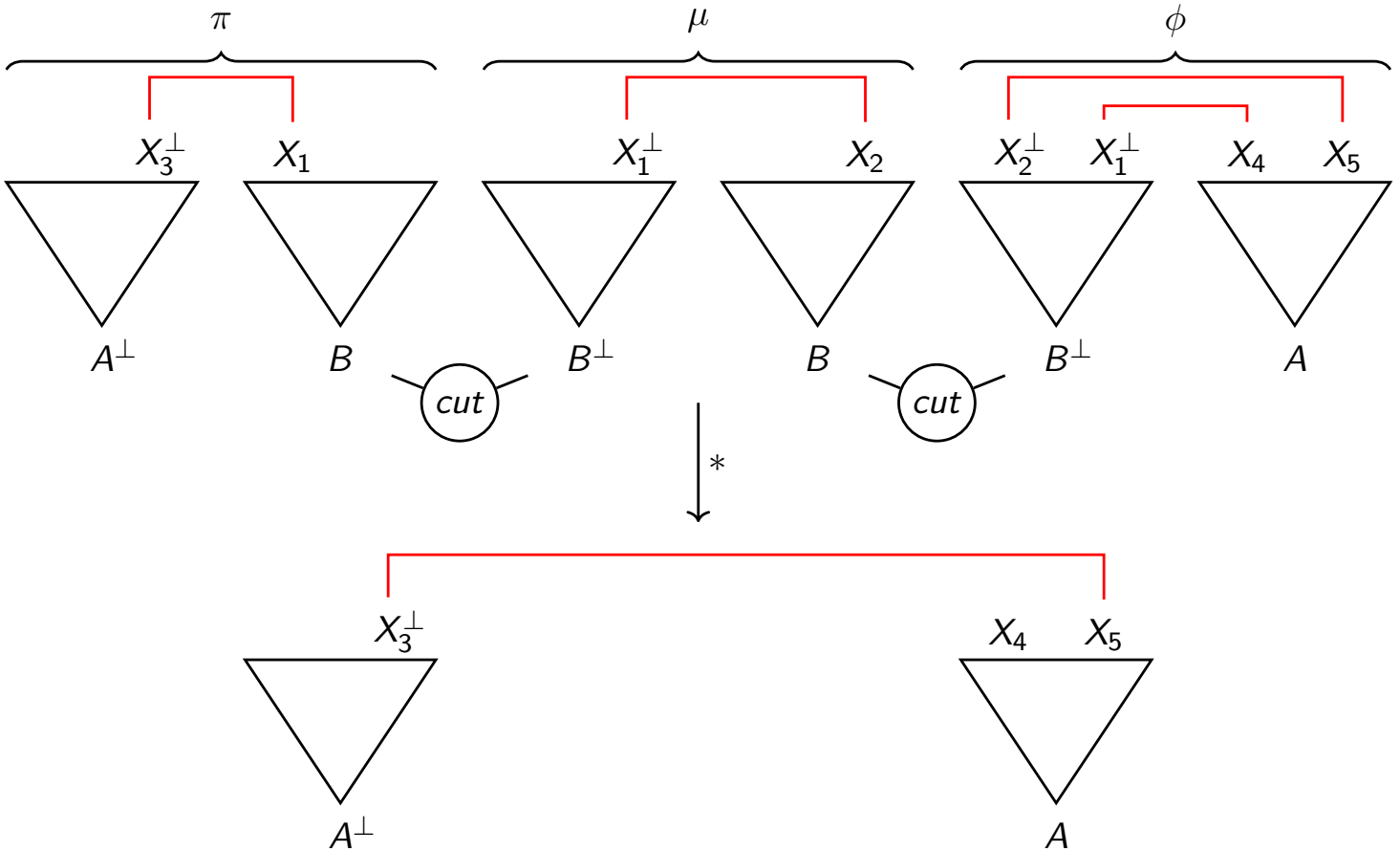
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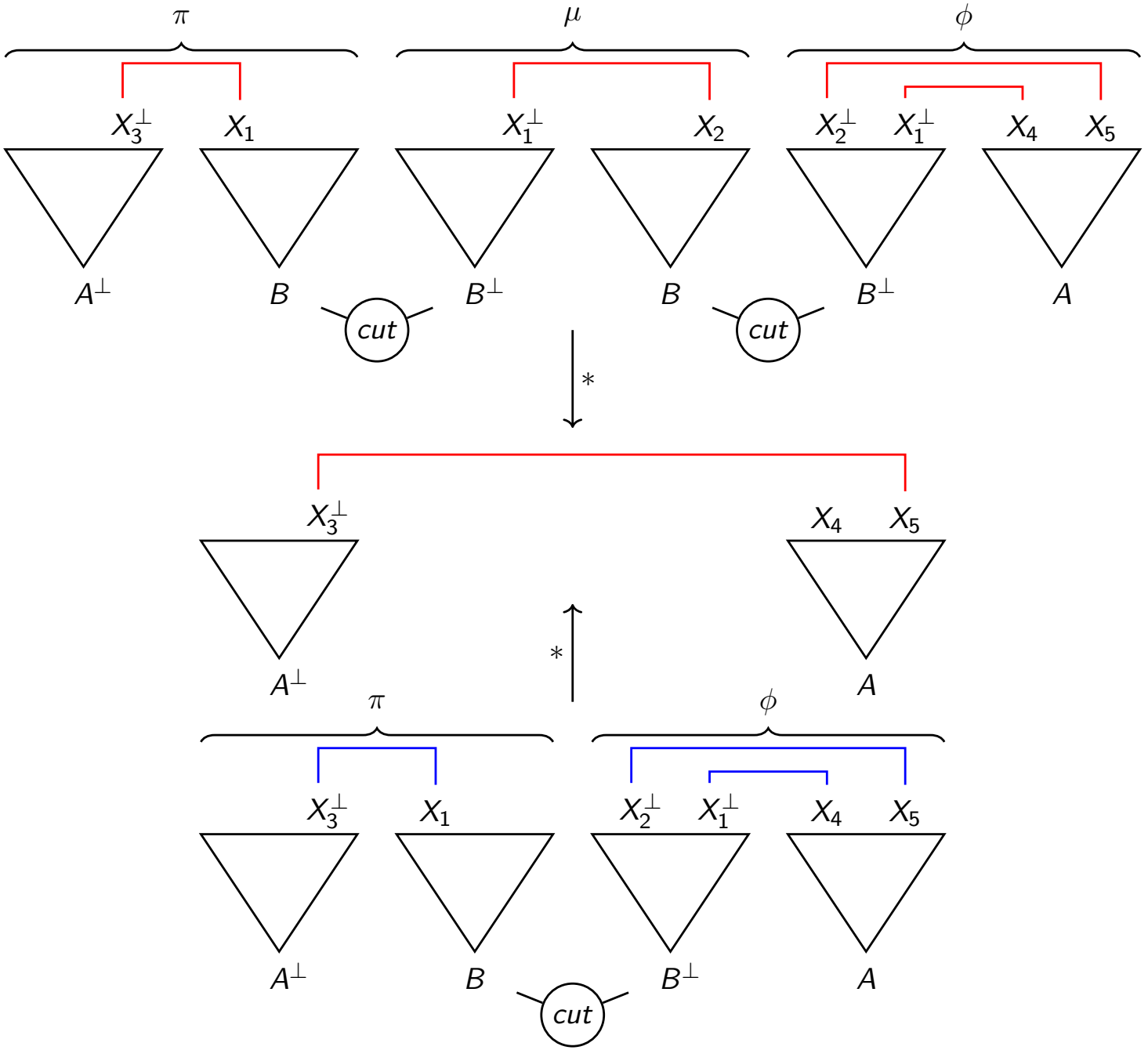
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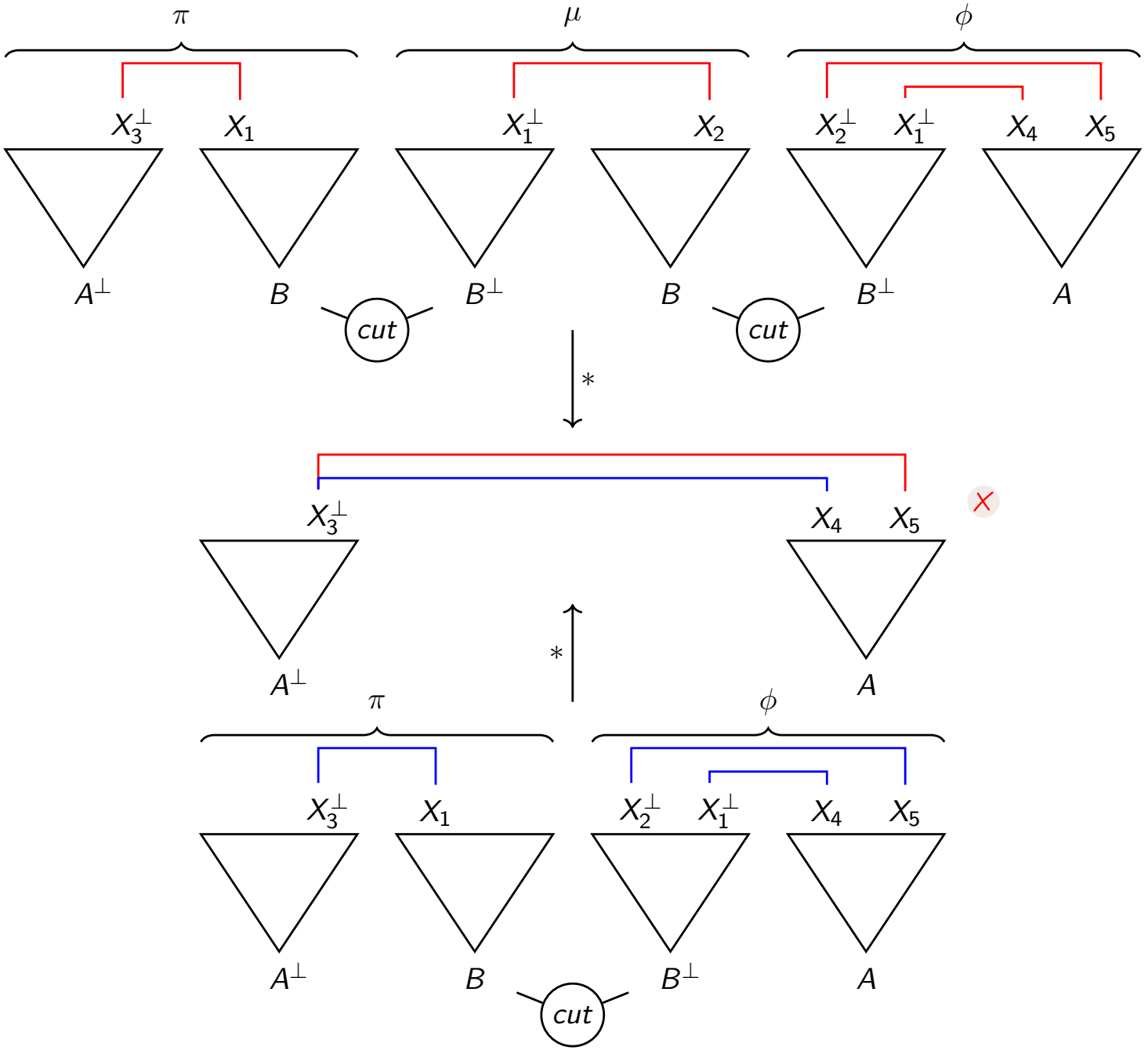
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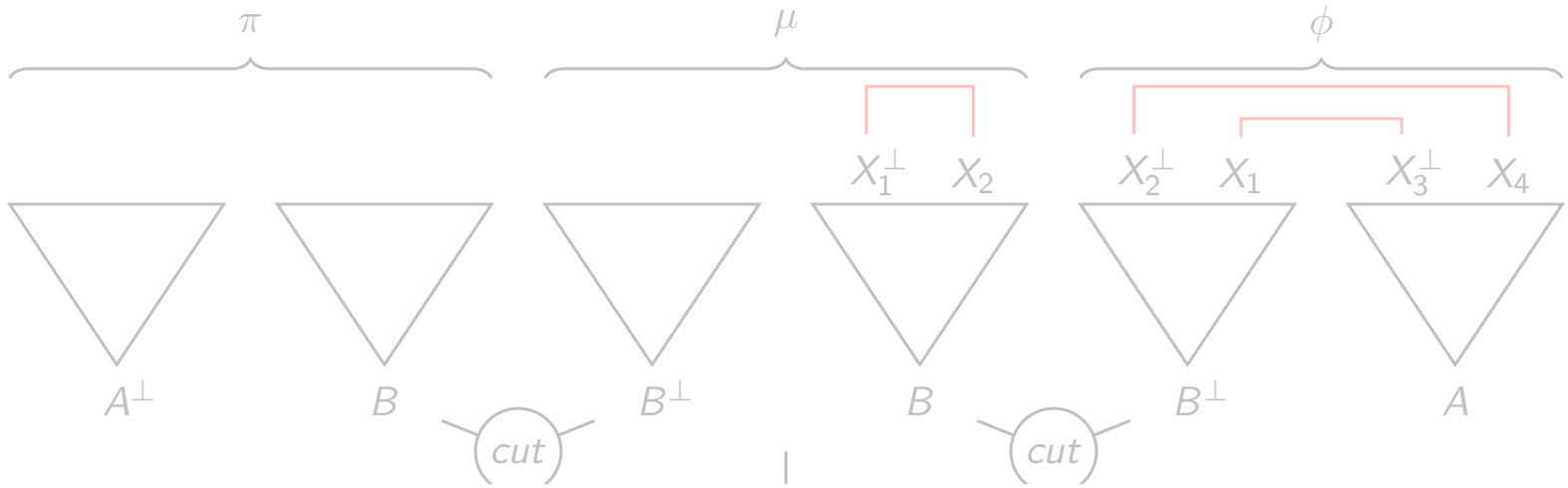
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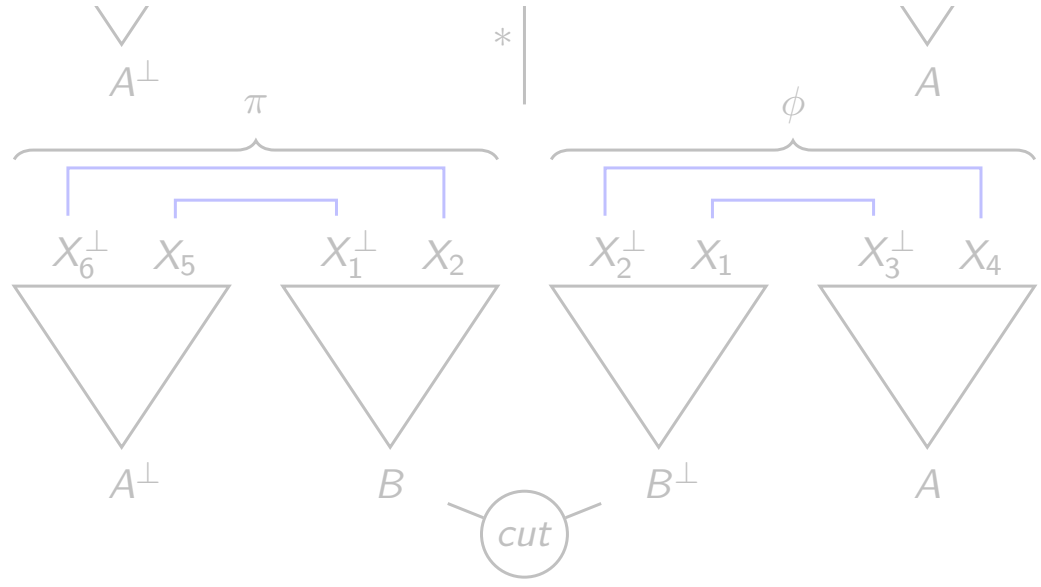
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- **Proof-Relevant Interpolation splits idempotents**: quite remarkable
  $\longrightarrow$ something *deeper / more general* behind this?
  just not too many ways to "logically" break a proof in MLL?
  (at least points to good properties of the inference algorithm)
  $\longrightarrow$ other logics where interpolation splits idempotents?
- **Extension** to LL?
  $\longrightarrow$ unit-free MALL seems easy, with units becomes way harder
  $\longrightarrow$ MELL (without *mix*) also harder
  $\longrightarrow$ quantifiers may be okay
- **Characterization** of idempotents / retractions ?



Thank you for your attention!



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